Received: 2 October 2022 | Revised: 13 February 2023 | Accepted: 12 March 2023

DOI: 10.1002/bit.28384

ARTICLE

BIOTECHNOLOGY BIOENGINEERING WILEY

Fully integrated downstream process to enable next-generation manufacturing

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Abstract

Next-generation manufacturing (NGM) has evolved over the past decade to a point where large biopharmaceutical organizations are making large investments in the technology and considering implementation in clinical and commercial processes. There are many well-considered reasons to implement NGM. For the most part, organizations will not fund NGM unless the implementation benefits the funding organization by providing reduced costs, reduced time, or additional needed capabilities. Productivity improvements gained from continuous purification are shown in this work, which used a new system that fully integrates and automates several downstream unit operations of a biopharmaceutical process to provide flexibility and easy implementation of NGM. The equipment and automation needed to support NGM can be complicated and expensive. Biopharmaceutical Process Development considered two options as follows: (1) design its own NGM system or (2) buy a prebuilt system. PAK BioSolutions offers a turn-key automated and integrated system that can operate up to four continuous purification stages simultaneously, while maintaining a small footprint in the manufacturing plant. The system provides significant cost benefits (~10× lower) compared with the alternative—integration of many different pieces of equipment through a Distributed Control System that would require significant engineering time for design, automation, and integration. Integrated and Continuous Biomanufacturing can lead to significant reductions in facility size, reduced manufacturing costs, and enhanced product quality when compared with the traditional batch mode of operation. The system uses new automation strategies that robustly link unit operations. We present the optimized process fit, sterility and bioburden control strategy, and automation features (such as pH feedback control and in-line detergent addition), which enabled continuous operation of a 14-day end-to-end monoclonal antibody purification process at the clinical manufacturing scale.

KEYWORDS

bioprocess, downstream, integrated system, intensification, NGM

1 | INTRODUCTION

Continuous manufacturing processes have gained significant interest from biopharmaceutical companies in recent years due to their potential to provide large productivity increases over traditional batch production modes (Arnold et al., 2019; Konstantinov & Cooney, 2015; Pollock et al., 2017; Zydney, 2016). These productivity increases allow a company to produce more biopharmaceutical product from an existing manufacturing facility or build new facilities with significantly reduced footprint compared to traditional designs (Croughan et al., 2015).

Progress on upstream continuous processes has been widely published and presented at industry conferences. Companies have reported titers for perfusion processes reaching 2.3 g/L/d and beyond at scales over 500 L (Coffman et al., 2021). This higher cell culture productivity results in a purification bottleneck for traditional batch purification processes. Perfusion cell culture volumes are typically on the order of 1.5 cell culture vessel volume per day over 30 days and have viable cell densities and titers that vary over 10× across this duration. A traditional 4 × 12,000 L facility can now be replaced with a 5 × 2000 L facility that outputs the same product mass, but at a cost of over 6× more cell culture volume that must be forward processed (Arnold et al., 2019). Purification facilities are not equipped with vessels to store the entire batch volume and, therefore, at a minimum, must process the material through the purification unit operations in multiple sub-batches (Jungbauer, 2013).

Continuous purification unit operations have matured in recent years and technologies now allow a full end-to-end continuous process without the need for large hold vessels (Rucker-Pezzini et al., 2018). These technologies eliminate the challenge previously presented by large perfusion harvest volumes. In a continuous purification process, each unit operation in the process is connected in series and operated simultaneously. As the material is continuously forward processed from one step to the next, there is no need to hold large intermediate volumes in tanks. Furthermore, the purification process keeps pace with the perfusion bioreactor and the full batch of drug substance is purified within as little as 4–12 h of completion of the cell culture operation (Coolbaugh et al., 2021; Gomis-Fons et al., 2020).

Productivity improvements gained from continuous purification extend beyond those achieved with perfusion cell culture processes (Klutz et al., 2015). Regardless of cell culture method, continuous purification offers benefits of reduced equipment size and footprint, reduced capital and consumable costs, and decreased run duration. Facility footprint is significantly reduced by elimination of product hold tanks and the need for utilities to clean and sterilize stainless steel piping systems. Low flow rates allow for smaller piping diameters that can be provided by single-use tubing instead of stainless-steel piping. Advanced automation allows processes to operate all steps 24/7 without intervention compared with one step performed per shift, greatly improving utilization of existing facility and personnel resources. Smaller chromatography columns are cycled continuously throughout a run, reaching more than 35 cycles compared with three to five cycles on a larger column for a traditional batch process (Bisschops et al., 2009). As a result, resin utilization is maximized and resin costs are reduced (Ötes et al., 2020). Although the capture chromatography step sees the most consumable cost reduction in a continuous purification process, labor and time savings for an end-to-end continuous process are notable (Mahal et al., 2021).

Companies are working together to develop strategies for end-to-end continuous purification processes (Erickson et al., 2021). Some processes have been published previously by our group and others (Coolbaugh et al., 2021; Feidl et al., 2020; Godawat et al., 2015; Levinson, 2016; Rucker-Pezzini et al., 2018; Schwarz et al., 2022; Steinebach et al., 2017) that include processes up to 25 days in duration and a demonstration of bioburden control and product quality. Considerations on scale-up and mixing have been studied (Anderson, 2001) and need to be included. Even so, significant opportunities for process maturity exist for certain unit operations, equipment hardware, and automation. Here we present the latest generation of continuous purification processes with an off-the-shelf continuous purification system that includes single-use flow kits compatible with good manufacturing practice (GMP) operation, advanced control strategies, and supervisory control and data acquisition (SCADA).

In this work, we demonstrate a 14-day continuous monoclonal antibody (mAb) purification process for a 50 L perfusion bioreactor. Bioburden was controlled within specifications for the duration of the process. A single automated system with four flow kits installed was used to perform the following steps: (1) harvested perfusion material was processed through a dual-column Protein A capture chromatography step, (2) a low pH virus inactivation step, (3) a depth filter, (4) a 0.2 µm sterile filter, and an (5) anion exchange membrane. The process performed did not require a cation exchange chromatography step to meet process impurity specifications. Ultrafiltration/diafiltration (UFDF) operations were performed separately after breaking down and setting up the same system in a UFDF setup.

As demonstrated in this work, continuous cell culture and purification processes have advanced beyond proof-of-concept studies. The significant advancements in off-the-shelf equipment now enable routine operation of continuous purification processes in pilot and GMP facilities. This equipment reduces footprint requirements for an end-to-end process by 75% and offers a level of automation that reduces hands on time by 90% compared to batch. Furthermore, we present low pH viral clearance data required to implement a continuous purification process for a GMP facility.

2 | MATERIALS

2.1 | Equipment and consumables

A pilot scale continuous purification system and four 500 mL single-use flow kits (γ-irradiated) from PAK BioSolutions were used to perform all purification process steps. MabSelect PrismA® and Sephadex G-25 Coarse® size exclusion chromatographic (SEC) resins were purchased from Cytiva. The three chromatography column housing units were Vantage L Laboratory Column VL 44 × 250 from MilliporeSigma. The Opticap Sterile 0.2 µm filter and the Mustang Q Membrane Capsule were purchased from MilliporeSigma and Pall, respectively. Tryptic Soy Agar plates were purchased from VWR.

2.2 | Buffers and solutions

The buffer system utilized for dual-column capture chromatography (MabSelect PrismA®) included six buffers: (1) 50 mM Tris, pH 7.4; (2) 50 mM Tris, 0.5 M NaCl, pH 7.4; (3) 25 mM sodium acetate, pH 3.6; (4) 100 mM acetic acid; (5) 1 N NaOH; and (6) 2% benzyl alcohol, 100 mM sodium acetate, pH 5.0. For the viral inactivation (VI) step, the solution system utilized 1 M acetic acid and 1 M tris base. All buffers were tested for bioburden and only qualified for use with negative results.

2.3 | Protein

The mAb (mAb1) used in the continuous process studies was a humanized IgG1 produced from Chinese hamster ovary (CHO) cells by AstraZeneca. This protein has a molecular weight of 146 kDa, as determined experimentally by mass spectroscopy, and has a theoretical isoelectric point of 7.15, as determined by amino acid sequencing.

The mAb (mAb2) used in viral clearance studies was a humanized IgG1 produced from CHO cells by AstraZeneca. The mAb2 intermediate (17 mg/mL, pH 4.5) purified by Protein A chromatography was used in the virus inactivation study as starting material.

3 | METHODS

3.1 | Process overview

The purification process for this study included an automated and integrated dual-column chromatography step (DCC) (Angarita & Morbidelli, 2015), leading into a continuous virus inactivation (VI) step and a final flow-through polishing AEX membrane step. Upstream and downstream of the DCC step were 10 L glass vessels to accommodate purification of 50 L of cell culture material per day. The process flow path, the connections between each unit operation in the process, and the automation to control and acquire the data from all unit operations in parallel, were performed with four 500 mL flow kits assembled with one pilot scale continuous purification system. The flow path of the process and the flow kits are shown in Figures 1 and 2, respectively. The 50 L per day of harvested material was combined from several different small scale perfusion bioreactors that had run before this study, in several bags. The order of the bags was chosen and connected to the downstream equipment to simulate the titer profile of a typical perfusion bioreactor process. The downstream process parameters are summarized in Table 1.

The 500 mL flow kits are three-dimensional-printed and include sensors (pH, conductivity, UV280nm, UV300nm, pressure), in-line mixers for buffer addition (pH titration, conductivity adjustment, in-line dilution), a vented mixing vessel, and a sample port (Figure 2).

3.2 | Sterilization and sanitization pre-run

Bioburden control of the continuous purification process required a combination of sanitization and sterilization methods. The flow kits and associated tubing assemblies were provided gamma-irradiated. Chromatography columns (MabSelect PrismA® used for the capture step and an SEC column used as a packed bed reactor for residence time during the VI step) were sanitized with 1 N sodium hydroxide during the automated start-up sequences, along with the associated flow paths that included pH, conductivity, and UV instrumentation. The 0.2 µm sterile filter and the Mustang Q membrane were sanitized offline per the vendor specifications with 1 N sodium hydroxide before use. Glass vessels used as product collection tanks, situated after the perfusion bioreactor and capture step, were autoclaved with tubing. Consumable components such as bags and tubing were γ-irradiated or autoclaved. Sanitary quick connects were used to make any connections that were not sanitized before use.

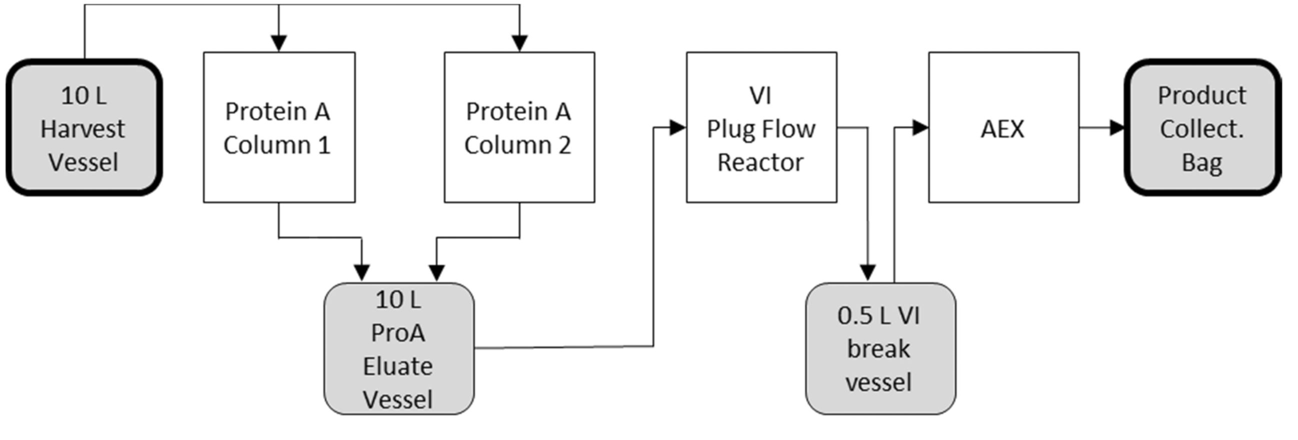

FIGURE 1 Process flow path in the continuous purification system. VI, viral inactivation.

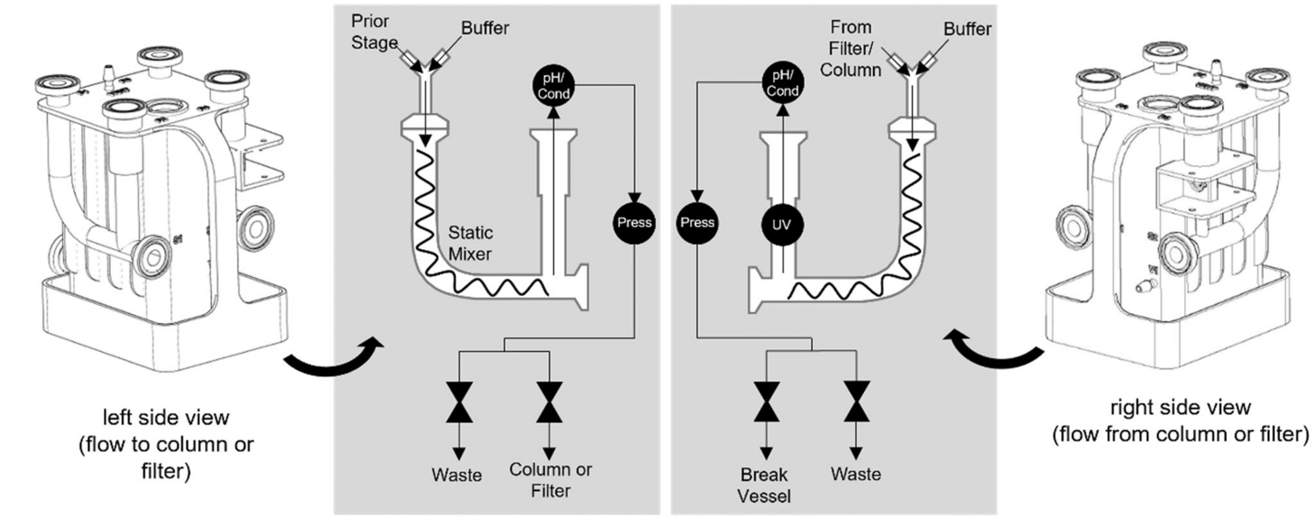

FIGURE 2 Flow path of the single-use flow kit for individual process steps (chromatography, virus inactivation, membrane, filtration). Includes, UV, pH, conductivity (Cond), and pressure (P) instrumentation.

TABLE 1 Process parameters.

	Protein A	VI packed bed reactor	AEX Mustang Q membrane	0.2 µm filter
Capacity	60 g/L	NA	150 g/L	1000 L/m^2
Size	4.4 × 7.7 cm (dual column)	4.4 × 20 cm	780 mL	0.059 m^2
Flow rates (mL/min) (min/target)	Load: 0–76	4.6/6.6–7.7/23.1[a]	0–50	0–50
Low pH (min/target)	NA	3.4/3.5/3.6	NA	NA
Neutralization pH		7.0/7.2/7.4		
Residence time (min) (min/target/max)	NA	5/15–17.5/25[a]	NA	NA

Abbreviations: NA, not available; VI, viral inactivation.

[a]Residence time target was changed from 15 to 17.5 min for VI Subrun 3.

3.3 | Cell culture

Several 3 L perfusion bioreactors were used to generate recombinant IgG via a CHO cell line. These individual 3 L perfusion bioreactors were harvested using Tangential Flow Filtration as a cell retention device, from which they were combined into 50 and 100 L bags according to their titers. The titer profile for the 14-day study (Figure 3) was mimicked by ordering the bags based on their titer to represent a typical 50 L perfusion bioreactor process. These bags were then welded to the flow path and pumped at a rate of 50 L/day into a 10 L glass vessel before the DCC step. The titer of the material used ranged from 0.2 to 2.34 g/L and totaled 490 g of harvested mAb.

3.4 | Dual column chromatography for Protein A capture

The target molecule was captured with two 4.4 cm diameter × 7.7 cm height chromatography columns using MabSelect PrismA® resin at a capacity of 60 g/L. The columns for the capture step were operated using a dual-column strategy, which operated in a typical bind and elute methodology, enabling a continuous load and discontinuous product elution. The size of the columns is optimized for protein mass and process time, resulting in higher resin utilization. Parallel column operation maintains one of the two columns in the load phase at all times. Although one column is in the load phase, the other column cycles through the process equilibration, wash, elution, strip, and sanitization phase buffers. The load time is never shorter than the sum of the nonload phase times, the "buffer phase duration." The columns were loaded to 60 g/L every cycle; the load duration and frequency of elution varied with the change in titer (load volume was manually input into the system's human machine interface [HMI] daily). The column was sized such that it took the same duration to reach maximum capacity on the highest titer day as the buffer cycle duration. This corresponds with the shortest load phase and, as a result, sets the minimum allowed column volume (CV) for the process. The elution collection criteria was fixed at five CVs and collection began after the completion of the first CV of the elution phase. The eluate was collected in a 10 L glass vessel downstream of the two columns.

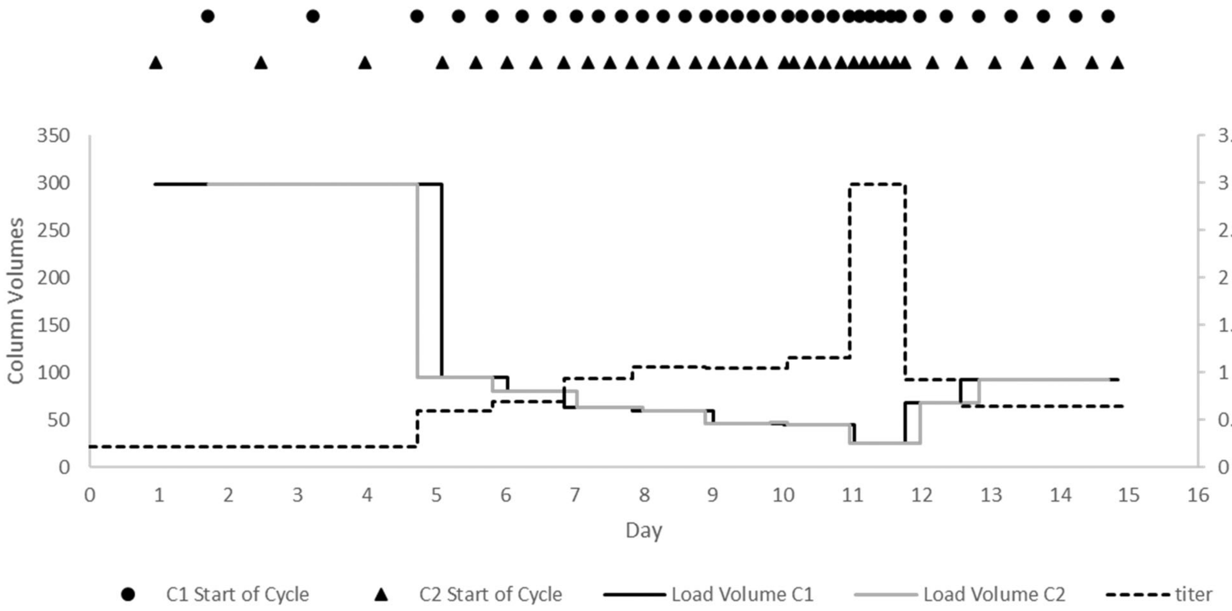

FIGURE 3 Protein A load volume and titer for the duration of the run. Black circles and triangles represent the start of the cycles for Column 1 and 2, respectively.

3.5 | Virus inactivation

The VI strategy implemented in this study used a plug flow reactor made from a 4.4 cm diameter × 20 cm height SEC column with Sephadex G-25 Coarse® resin to achieve a target residence time of 15 min (Arnold et al., 2019). Acidification was performed before the SEC column and neutralization was performed post column. The capture product was titrated to an acidification target of pH 3.5 with an allowable range of ±0.1. Acid was added in-line with the product line and the acidified product immediately flowed through the flow kit's static mixer and pH probe. A feedback proportional-integral-derivative (PID) controller is used to continuously adjust the acid pump flow rate to ensure the pH is within the desired range. After the pH is within the acidification target range for an appropriate amount of time, the product is directed through the SEC column which was equilibrated at pH 3.5. Post SEC column, an in-line addition of base is continuously fed to neutralize the acidic product to a target of pH 7.2 with an allowable range of ±0.2. The VI procedure during this process began on Day 8 of the 14-day run. The start day for this step is determined based on (1) product quality impact from hold time at capture product conditions and (2) system flow rate lower limit to allow VI to start and run for extended duration. For this study, VI was run in three sub-batches. For a future optimized process, we aim to operate without pausing.

Xenotropic murine leukemia virus was used in the spiking experiments for both batch and continuous processes. The virus content was using plaque assay for infectivity and polymerase chain reaction (PCR) for viral genome copy number.

3.6 | Anion exchange chromatography

The product from VI step flowed through a filter train comprised two Opticap® Sterile 0.2 μm filters on either side of a Mustang Q® Membrane Capsule. This filter train was prepped offline, before its connection to the system according to the vendor specifications. This prep work was done the day before the filter train was connected to the system (Day 7). The filter train was connected on Day 8; this timeline coincided with the start of the VI step. From here, material was collected in a final 50 L product collection bag. This final polishing chromatography step was performed with a single-use membrane in a flowthrough mode. The membrane was not reused. The 0.2 μm filter downstream of the Mustang Q membrane ensured that no potential bioburden reached the final product collection bag.

3.7 | Bioburden sampling and testing strategy

Samples were taken from four points in the process: (1) the cell culture harvest 10 L vessel, (2) Protein A product 10 L vessel, (3) VI product flow kit vessel, and (4) final product collection bag. Samples were taken aseptically using 10 mL syringes, about 24 h apart. The samples for testing bioburden were taken and plated onto culture plates under a biosafety cabinet and stored inside of an incubation chamber running at 35°C. The plates were monitored every day for 72 h to inform sterility of the system. The remaining samples were stored at 2–8°C until they were ready to be submitted for analysis (see Section 3.8 for more details).

The Protein A bindable testing was done to record the daily titer at each stage. The size exclusion-high performance liquid chromatography (SE-HPLC), host cell protein (HCP), and DNA assays tested the impurity level at each stage and therefore show how effectively each unit operation removed impurities.

3.8 | Analytical testing

The Protein A bindable testing was done to record the daily titer at each stage. The SE-HPLC, HCP, and DNA assays tested the impurity level at each stage and, therefore, show how effectively each unit operation removed impurities.

3.8.1 | SE-HPLC

SE-HPLC was performed using a Waters 515 pump and 2487 Dual Absorbance Detector with a TSK Gel G3000 SWXL column (300 mm × 7.8 mm, Tosoh Biosep). The separation was performed at a flow rate of 1.0 mL/min, using 100 mM sodium phosphate buffer, pH 6.8, with UV detection at 280 nm. The results were reported as percentage monomer and high molecular weight peaks.

3.8.2 | CHO DNA real-time PCR assay

Sample DNA was extracted using a DNA Extractor Kit® from Wako. The assay was performed in 96-well plate format using primer set specific for the Chinese hamster genomic DNA with 40 cycles of PCR and monitored using SYBR Green for quantitation relative to a standard curve.

3.8.3 | Gyros CHO HCP assay

Samples and controls were appropriately diluted and tested in microplate format using Gyrolab instrumentation. The assay included biotinylated sheep anti-CHO HCP capture antibody and fluorescent labeled sheep anti-CHO HCP detection antibody and was quantitated using a standard curve.

3.8.4 | HPLC Protein A bindable assay for mAb concentration

Sample was injected onto a Protein A immobilized PA Immuno-Detection Sensor Cartridge (2.1 mm × 30 mm) in phosphate-buffered saline (PBS) buffer. The bound IgG was eluted with 0.1% phosphoric acid in PBS and detected using UV absorbance at 280 nm. The Protein A bindable IgG content was quantified using a standard curve.

4 | RESULTS AND DISCUSSION

This study successfully demonstrated a next-generation continuous purification process, that ran for 14 days at a rate of 50 L of cell culture harvest material per day. Bioburden control was maintained for the duration of the run and product quality results were within acceptable limits. Advanced automation enabled tight control of the process to meet specifications. Basic automation included: pump and valve control, UV-based elution collection, min/max flow rate control, and automated waste diversion. In addition to this, comprehensive automation included control of vessel levels to setpoints, control of pH to targets with inline titration, and overall mass flow control. Harvest vessel and VI break vessel volumes were controlled to setpoints using PID control on the Protein A feed pump and VI feed pump, respectively. The Protein A eluate vessel was controlled within high and low limits, which triggered downstream steps to pause or resume. The pH was controlled with dynamically tuned PID loops on the titrating pumps and had timely waste diversion for out of specification material. The mass flow rate of the process was controlled to match the VI residence time target and is described in detail in later sections. Automation control strategies also enabled break vessel volumes between 150 ml and 10 L (20-fold or greater reduction compared with batch). Facility footprint of the process was reduced by 75% compared with the batch process, primarily due to the reduction in vessel volumes and ability to perform all three process steps in a single system, which occupies similar space of a single chromatography skid. Setup time was 1.5 days and was followed by 14 days of hands-free operation with the exception of daily sampling (90% reduction in operator hours). These outcomes demonstrate the robustness of this commercially available automation system for continuous purification processes.

4.1 | Bioburden control

A traditional batch purification process for a mAb is not considered aseptic and operates as low-bioburden. The guidance commonly used sets a limit at 10 colony-forming units (CFU)/mL.

Bioburden control was demonstrated for the duration of the 14-day run. Samples were taken daily from the harvest break vessel, Protein A product vessel, VI-neutralized product vessel, and AEX final product. None of the sampling points tested positive for bioburden, except on Day 7, when 1 CFU/mL was observed for the Protein A product vessel (below the acceptable criteria of <10 CFU/mL). This hit was attributed to intentionally opening the Protein A product vessel to fix an incorrectly installed tubing assembly on the autoclaved vessel. The bioburden objectives for this process were clearly met. The 1 CFU/mL sample from the Protein A elution product vessel was cleared by implementing a bioburden mitigation strategy. This strategy consisted of six important steps: (1) pausing the Protein A step, (2) removing the contaminated vessel from the system, (3) autoclaving a new vessel, (4) filtering material into the new vessel under aseptic conditions, (5) integrating the new vessel into the process, and (6) flushing lines with sanitization buffer before restarting the Protein A step. No bioburden was detected after the implementation of this strategy. Successful implementation of this mitigation strategy demonstrates the ability to clear bioburden from an ongoing process without significant process disruption, a necessity to save a batch when operating a continuous operation.

4.2 | Capture

Seventy Protein A capture chromatography cycles were successfully run during the 14-day process. The Protein A capture step began automatically once the harvest vessel reached a set point volume of 2 L. The capture step operated for the duration of the run.

The capture step was operated under a constant mass load principle, targeting the resin capacity of 60 g of mAb per liter of resin: the load volume onto the column changed as the titer varied throughout the run to ensure that the grams of mAb loaded was the same for each cycle, as illustrated in Figure 3. Load volume decreases as titers increase until the middle of the run. Load volume then increases again as the titer drops towards the end of the run. This strategy ensures a constant concentration downstream of the Protein A capture step, which greatly simplifies the downstream process dynamics. The SD for the Protein A concentration over the duration of the run was 1.4 g/L. With constant downstream concentration, the virus inactivation step does not need to account for large mAb concentration changes that can affect the titration profile. Additionally, polishing chromatography step throughput can be calculated from volumetric throughput and the known concentration. The capture step was triggered to stop once the harvest vessel dropped to a low level due to lack of flow from the cell culture feed.

Chromatogram overlays of the 35 cycles performed on each of the two capture columns were generated for analysis (data not shown). Constant peak height and position are observed across all cycles. Tailing was noticed on some cycles, especially towards the end of the run. Analysis of conductivity traces showed a change in transition curve profile for these cycles which indicates a change in bed structure rather than a change in impurity profile or issue with the UV meter. The root cause was likely air on the columns. This can easily be mitigated for future processes by installing a bubble trap before the Protein A step, which was not in place during this run.

4.3 | Virus inactivation

The continuous low pH virus inactivation step was successfully controlled to meet pH and residence time specifications. The virus inactivation step was performed in three parts by starting and pausing this step three times during the 14-day process. The Subruns 1 and 2 were target to 15 min residence time, whereas the Subrun 3 target 17.5 min. These pauses were required to account for the change in rate of product elutions from the upstream Protein A step and maintain a constant residence time across the virus inactivation packed bed reactor. The virus inactivation step was triggered to automatically start when the Protein A product vessel of 10 L filled to a high target volume of 7.5 L. During the run, the virus inactivation step drained the vessel until it reached a low target volume of 1.5 L, at which point the virus inactivation step automatically paused until the high target level was reached again. These trigger levels were manually changed through the HMI throughout the run to allow for the trigger events to coincide with an operator onsite. This was due to an abundance of caution and is not required for the process to operate. Each of the three virus inactivation subruns lasted ~1 day.

The virus inactivation step used an SEC column to provide the required inactivation residence time. The SEC column models a plug flow reactor and this choice benefits from the abundance of industry knowledge around column packing and qualification. Importantly, the resin is inert for protein and virus binding and the pore size is smaller than the virus diameter, resulting in no virus mass transfer between mobile and stationary phases. Virus flows only in the column's void space, as experimentally demonstrated later in this report (Figure 8). The residence time of the column was experimentally determined and defined as the front edge of the peak. Defining the front edge of the peak as residence time for process control is most conservative, because any peak broadening or tailing would add time to enhance virus inactivation. This is also reflected in consistent volumetric breakthrough profiles at operating velocities (Figure 7), confirming that diffusion is negligible, allowing simple calculation to confirm residence time for any particular flow rate. This simplicity is not seen with tubular flow reactor designs, which rely on dean vortices for mixing, as peak broadening is dependent on flow rate. This results in residence times that are challenging to calculate with changing process flow rates (Amarikwa et al., 2019). Other options, such as alternating tank VI, were not considered due to the increased space, equipment, and automation requirements.

The column was first equilibrated at pH 3.5. The pH of the capture product was titrated inline down to a pH of 3.5 with 1 M of acetic acid, before passing through the SEC column (Figure 4a). After exiting the SEC column, pH was titrated inline up to a target pH of 7.2 with 1 M tris base. Inline pH adjustments were performed using feedback control. The pH was well controlled for and was mostly maintained within the specified ranges of ±0.1 and ±0.2 for the acidification and neutralization, respectively. Any disturbances outside of the specified ranges were diverted to waste until pH returned within specification and remained stable for 2 min. The pH data are shown in Figure 4.

Residence time through the packed bed reactor was also controlled within the allowable ranges through pump feedback control. Feedback control, rather than a fixed flow rate, was required, because the virus inactivation feed pump must maintain flexibility to control the virus inactivation break vessel volume. This strategy is required when running a truly integrated process as fixed flow rates on any step will inevitably result in vessel over/underflow events due to imperfectly calibrated pumps. Details of this strategy are explained later in the Advanced Automation Control Strategies section.

4.4 | Polishing

A flowthrough anion exchange membrane (Pall Mustang Q) polishing step was performed for this run. A second polishing step was not required to meet product quality specifications.

The filter train, which consisted of the AEX membrane and two 0.2 μm filters (pre- and postmembrane), was assembled accordingly

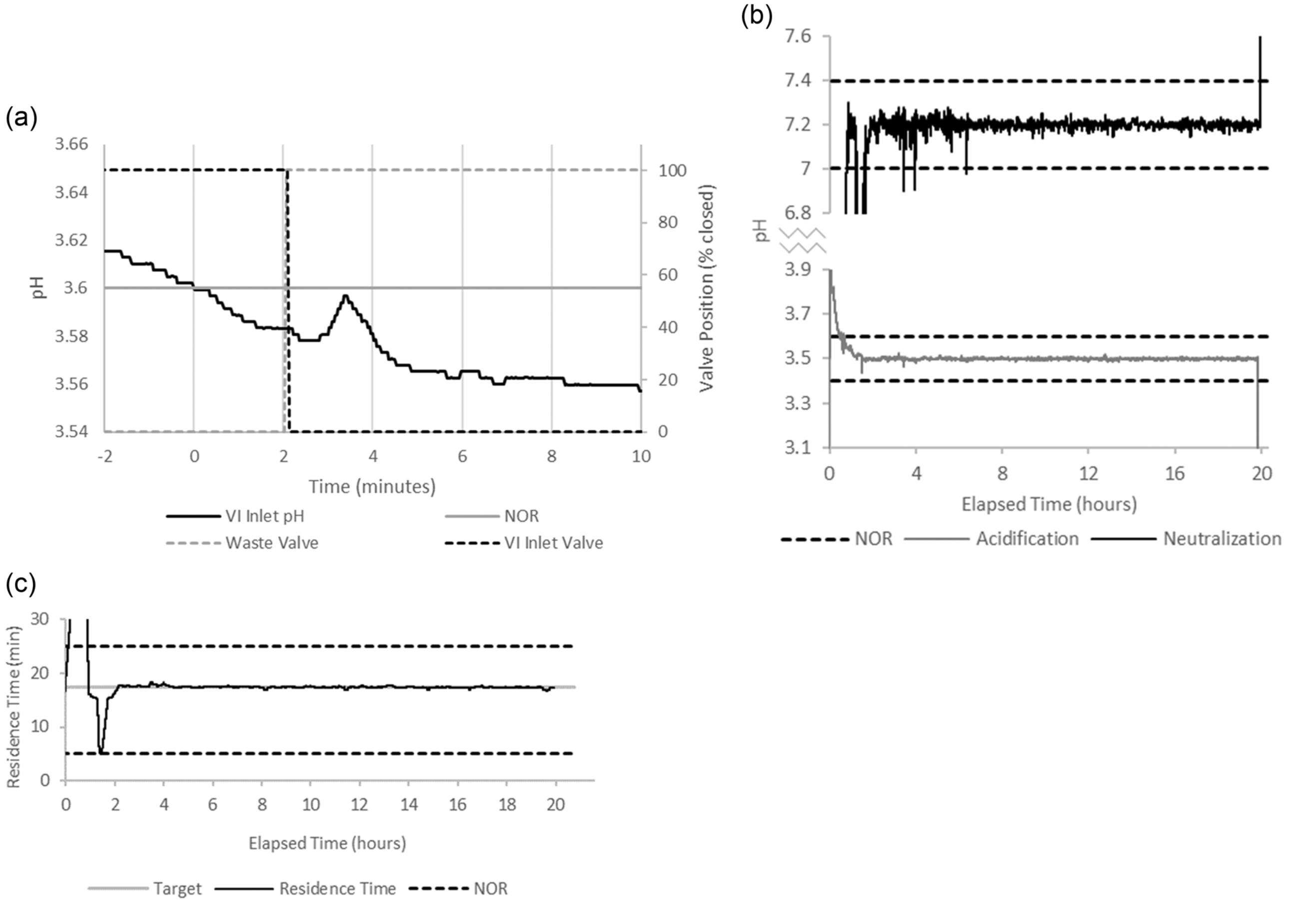

FIGURE 4 Virus inactivation pH trends. (a) Valve switch to in-line with process during startup. (b) pH trends for Subrun 3. NOR is normal operating range defined by the process description. (c) Virus inactivation residence time trend during Subrun 3.

to the filter's area and capacities to allow replacement of all at the same time. During process fit, it was determined the number of times the train would need to be swapped. The assemblies were prepared and connected for an easy valve switch when the swap is necessary. The throughput was automatically determined by the system, which alerted the user when a swap was required.

The filter train was swapped once during the process, although per the process fit, two swaps would be necessary. Unfortunately, the coronavirus disease pandemic impacted global supplies and limited the availability of single-use materials such as filters and membranes; thus, only one swap was able to be performed. This resulted in an intentional overloading of the second Mustang Q to prevent cutting the run short. The first train was loaded to 150 g/L of membrane (14.2 L), whereas the second was loaded to 324 g/L (30.4 L). Aside from the consequence of limited membrane area, the polishing step ran as expected.

4.5 | Product quality

The continuous purification process performed as expected and demonstrated acceptable product quality results. DNA, HCP, and monomer purity measurements were performed. Samples were taken from the harvest vessel, Protein A product vessel, virus inactivation product flow kit, and the final collection bag. The final collection bag contained all the material collected from a single filter train. Product was diverted to a new collection bag when the filter train was swapped. The final AEX product was within acceptable ranges for percent monomer and HCP on both days measured. DNA levels were acceptable on Day 9 but out of range on Day 14. This was attributed to the intentional overloading of the second AEX membrane as described above. Overall, product quality was comparable to the batch downstream process, with the exception of the Day 14 AEX DNA data point. Results can be seen in Figure 5.

4.6 | Advanced automation control strategies

A key objective of this work was to evaluate the ease of integrating multiple unit operations, which requires complex automation strategies. Unit operation integration was handled completely by the system's automation, which removed the burden of in-house automation development and implementation. With this system, unit operation integration is mediated by system-monitored break vessel

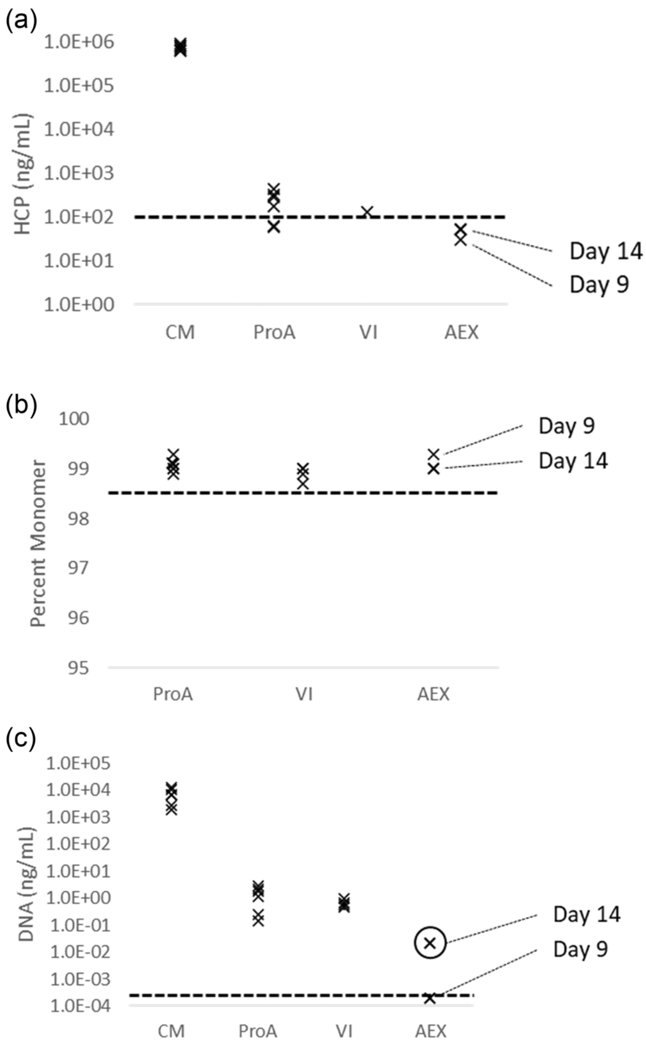

FIGURE 5 Product quality results from each sampling point across the duration of the run. Black dashed line indicated the acceptable cutoff for the final AEX product. Each data point is from a different day in the process. AEX final product sample points are labeled with day of sample collection. (a) Host cell protein (HCP) titers in ng/mL (acceptable cut-off below line). (b) Percent monomer (acceptable cutoff above line). (c) DNA titers in ng/mL (acceptable cutoff below line). Circled DNA data point on AEX step shows out of spec sample due to intentional overloading of AEX membrane.

levels. This run included (1) a harvest break vessel, (2) a Protein A capture product vessel, and (3) a VI product flow kit vessel. The harvest break vessel and the VI product break vessel were controlled to a level set point using PID feedback control over the vessel outlet and inlet pump, respectively. The Protein A product vessel used minimum, low, high, and maximum level criteria to trigger downstream pause/restart events. Overall, all vessels were controlled as intended. The significance of this control strategy is that it eliminates the concern of over/underflow of intermediate break vessels.

The Protein A elution vessel changeover performed on Day 9, as described in Section 4.1, resulted in a buildup of cell culture material in the harvest vessel, as the system was paused during this time while perfusion material continued to accumulate. The 10 L capacity of the harvest vessel coupled with its 50 L overflow bag provided ample time for an operator to swap vessels and restart the system without further incident.

A critical, yet often overlooked parameter in a continuous process is the mass flow rate of the overall process. Rather than set each unit operation to a specified flow rate and oversize break vessels to absorb differences, the system's automation actively controls the flow rate of each individual step and the system as a whole. This is achieved by controlling the last product pump in series to a flow rate set point, while regulating flow of upstream product pumps to maintain intermediate break vessel level (the Protein A capture step feed pump is an exception as it is controlled by the harvest vessel level as described previously). The break vessels controlled by upstream pump PID control allow for changes in the last product pump to cascade up the system. For example, an increase in the flow rate of the last product pump will drain the vessel immediately upstream. As the vessel drains, its feed pump will speed up to compensate, because it aims to maintain the vessel level target. This process continues to cascade up the system. Thus, as a result of the vessel level control, any change in flow rate of the last product pump results in a corresponding change in flow rate of the upstream product pumps. This control strategy allows for low break vessel volumes (150 ml for this scale – 50 L cell culture per day process).

The mass flow rate set point (set for the last product pump in series) is determined by the most critical process parameter. For this run, the most critical process parameter was determined to be the virus inactivation residence time. Therefore, the mass flow rate of the entire process was controlled such that the VI residence time was actively driven to a set point. As a safeguard, the total flow through the VI is limited so that it cannot result in residence times that fall outside the bounds of the user defined minimum or maximum. This is important during process perturbations or startup where the residence time may deviate away from target. A schematic of this process is displayed in Figure 6.

The following steps are performed to adjust the pump flow rates to drive the virus inactivation residence time to a set point. First, the total flow into the virus inactivation SEC column is calculated as the sum of the Protein A product feed (VI product pump) and the VI acid pump flow rates. The flow rate set point that corresponds to the virus inactivation residence time target is then subtracted from the total inlet flow, providing the flow rate error. The flow rate error is fed into a PID controller that controls the AEX product pump (the last product pump in series in this process). As this pump pulls from the VI break vessel, any change in this flow rate will immediately result in a change in VI product vessel level. The error of this vessel is fed into a PID controller that controls the VI product pump to maintain a constant VI vessel level. Thus, increasing or decreasing the flow rate of the last product pump in series results in a respective increase or decrease in the upstream VI product pump flow rate simply due to process dynamics from the PID control over the VI vessel level. The increase or decrease in VI product pump flow rate results in a respective decrease or increase in VI residence time through the packed bed. Finally, the VI acid and base pumps increase or decrease their flow rates so that the target ratio between F_{acid} and $F_{proA\ eluate}$, and between F_{base} and F are always met.

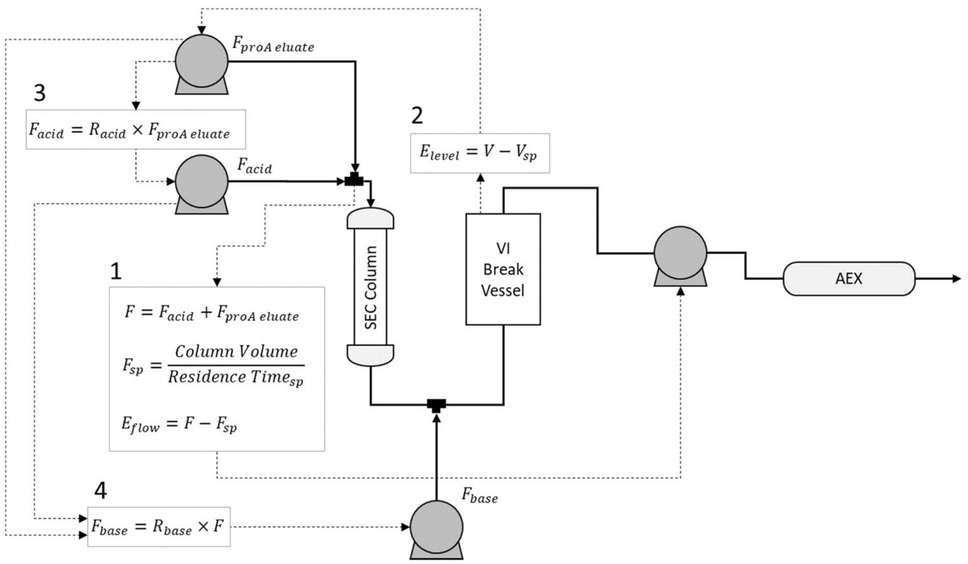

FIGURE 6 Schematic of mass flow rate control. (1) $F_{proA\ eluate}$, Protein A product material flow rate; F_{acid}, acid flow rate; F, total flow into VI SEC column; F_{sp}, set point flow rate; E_{flow}, error in target flow into SEC column. (2) E_{level}, error in target vessel level; V, vessel level; V_{sp}, set point vessel level. (3) R_{acid}, target ratio of acid flow rate to product flow rate. (4) F_{base}, base flow rate; R_{base}: target ratio of base flow rate to F.

The virus inactivation residence time for the third VI subrun is displayed in Figure 4c.

This series of independent control loops, connected only by the physical properties of the VI product vessel allows for cohesive mass flow control across unit operations. Additional unit operations can be added between the VI and AEX steps with this strategy. In this case, flow control cascades up through multiple break vessels.

The perfusion process required a break in mass flow rate at some point in the process. The perfusion rate must be set constant and for this run the VI residence time was controlled to a set point. Regardless of the accuracy of pumps, there will inevitably be a difference in mass flow rates between these two unit operations. The break in mass flow rate occurred between the Protein A chromatography step and the virus inactivation step. The difference in mass flow rate was absorbed by the Protein A product break vessel. This is the reason the Protein A product vessel operated under min, low, high, max control rather than continuous PID control and why the virus inactivation step ran as three separate sub-batches during this run. In future runs, the VI step will operate within a range of acceptable residence times, which will allow VI flow rates to adjust to match the changing output of the Protein A eluate vessel. This will enable this step to run continuously for the duration of the process instead of periodically stopping and restarting.

4.7 | Viral clearance data from continuous virus inactivation

To ensure viral clearance across the flow through virus inactivation SEC column, live virus spiking studies were performed. These studies aimed to (1) characterize the breakthrough profile of virus

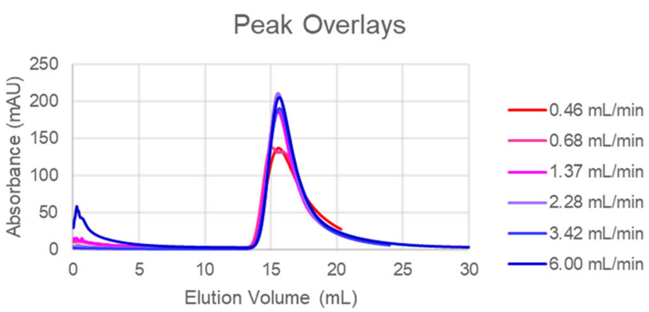

FIGURE 7 Peak overlays of all blue dextran elutions. Variance in peak height is due to variations in blue dextran spiking concentration and is irrelevant to the results.

and (2) to ensure inactivation kinetics occur in the same manner to batch inactivation.

Before testing the column with the virus-spike, the breakthrough profile of the column was characterized with blue dextran as a virus surrogate. Blue dextran is a large molecule with a molecular weight of ~2000 kDa (~10–20 nm diameter). It is commonly used to determine column void volume and is expected to behave similarly to enveloped viruses, which are typically larger than 50 nm (ICH Q5A, Food and Drug Administration, 1999). Blue dextran spiking studies were performed at various flow rates across the SEC column to characterize the breakthrough profiles of different column residence times. Data can be seen in Figure 7.

As can be seen from the results, breakthrough volume was independent of flow rate, with a standard deviation of only 0.004 CVs. This can be explained by the Van Deemter Equation:

$$\text{HETP} = A + B/u + (C_s + C_m)\cdot u,$$

where A is the constant eddy diffusion parameter, B is the diffusion coefficient in the longitudinal direction, and C_s and C_m are the resistance to mass transfer in the stationary and mobile phases, respectively. As there is no mass transfer of blue dextran from the liquid phase to the solid phase or vice versa, this portion of the equation can be ignored. The flow regime is also operated at a relatively large u, making the longitudinal diffusion part of the equation negligible as well. This eliminates any dependence of velocity on peak broadening. These results are significant, because they demonstrate a consistent breakthrough profile regardless of flow rate, which may need to be adjusted throughout the course of a continuous process. Therefore, breakthrough time across the column can be easily calculated from the current flow rate, CV, and void volume coefficient. This is not necessarily true of other methods of flow through virus inactivation that rely on Dean vortices for axial mixing.

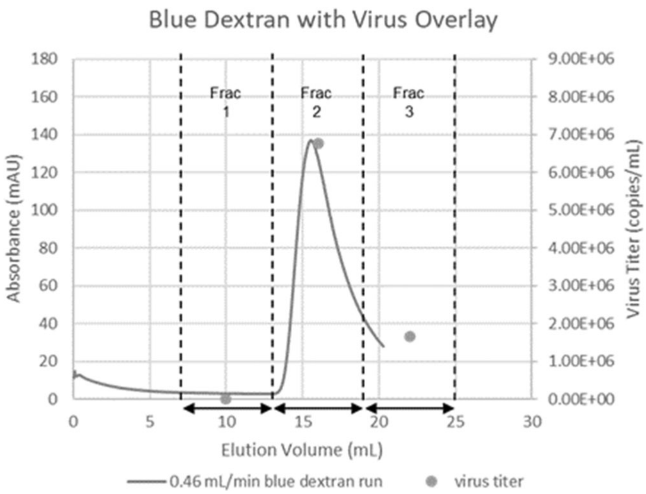

FIGURE 8 Breakthrough of XMuLV overlayed with a blue dextran run. Both data sets performed at 7.3 cm/h.

To demonstrate that live virus behaves similar to blue dextran, the column was spiked with inactivated xMuLV virus while running at a residence time of 30 min. Three 6 mL fractions of eluate were collected corresponding to before, during, and after the expected breakthrough of virus. Results can be seen in Figure 8.

The pre-breakthrough fraction contained 2.85×10^1 virus copies/mL, indicating little to no breakthrough as expected. The expected breakthrough fraction contained 6.78×10^6 virus particles and the post breakthrough fraction contained 1.65×10^6 virus particles. This follows a similar profile to the blue dextran elution profile and, most importantly, shows virus breakthrough does not occur earlier than blue dextran.

In addition to breakthrough characterization, virus inactivation was also evaluated at multiple residence times across the column. Virus inactivation was measured at residence times of 2, 10, 20, and 30 min. Log inactivation data is shown in Figure 9.

The log inactivation data show a similar trend to the batch inactivation data for the same molecule under the same conditions. This clearly shows there is no difference in the kinetics of inactivation between batch and continuous methods. The virus was inactivated by more than 4 $\log_{10}$ or 10,000-fold risk reduction withing 5 min in the continuous mode, supporting a 5 min residence time minimum.

5 | CONCLUSION

We have successfully demonstrated a next-generation continuous purification process for a 14-day operation with bioburden control. The PAK automated system for continuous purification provided the automation control, automation hardware, and single-use assemblies required. Advanced control strategies controlled flow from one step

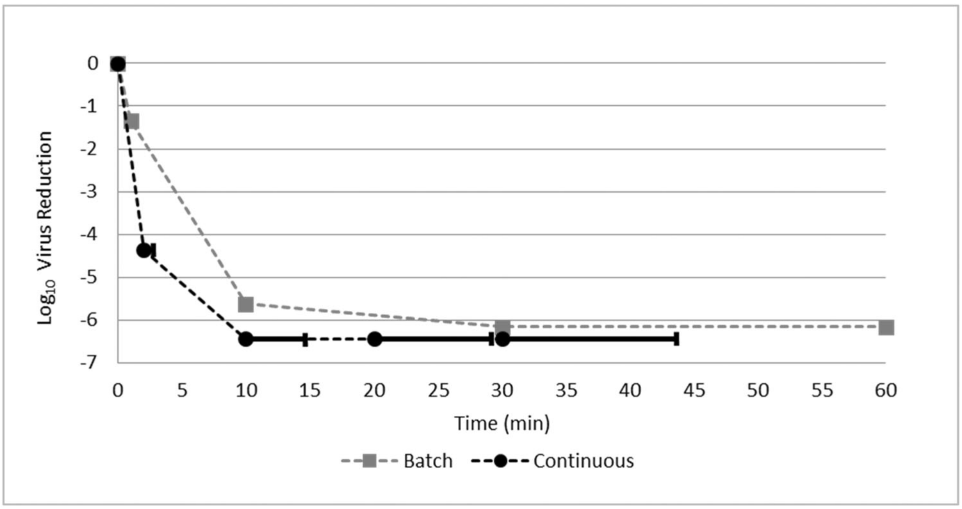

FIGURE 9 Log reduction values for both continuous and traditional batch modes of low pH virus inactivation. Both data sets performed at pH 3.6 with the same molecule. Dots on continuous data set indicate start of collection of eluate sample, whereas lines indicate the span of collection of the sample.

to the next and the mass flow rate of the entire process. The control parameters were continuously monitored and set to alarm when limits were reached. No additional automation control or SCADA system was required to perform this work. Automation used was 21CFR Part 11 compliant and all product contact materials met or exceeded USP Class VI specifications. The low pH continuous VI method used in this process was tested by off-line virus spiking studies and results are comparable to the batch process. The dual column operations performed in the continuous process offers a low regulatory impact to the traditional batch operation. Additional studies are required to confirm viral filtration efficacy at the reduced flow rates used in the continuous process. The technology presented enables implementation of continuous fully integrated processes at scales up to 200 L of cell culture material per day, which is a significant step towards GMP implementation of end-to-end continuous purification processes. Subsequent work to fit a three-column purification process has been initiated, in which two PAK systems can be synchronized to enable all the unit operation to run in concurrently (data not published yet).

AUTHOR CONTRIBUTIONS

Study conception and design: Irina Ramos, Joanna Pezzini, Nikunj Sharda, Kevin Hill-Byrne. *Data collection*: Ramon Villafana, Nikunj Sharda, Kevin Hill-Byrne. *Analysis and interpretation of results*: Irina Ramos, Joanna Pezzini, Kang Cai, Ramon Villafana, Nikunj Sharda, Kevin Hill-Byrne. *Draft manuscript preparation*: Irina Ramos, Joanna Pezzini, Kang Cai, Ramon Villafana, Nikunj Sharda, Kevin Hill-Byrne, Jon Coffman. All authors reviewed the results and approved the final version of the manuscript.

ACKNOWLEDGMENT

Not applicable

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Ramos, I., Sharda, N., Villafana, R., Hill-Byrne, K., Cai, K., Pezzini, J., & Coffman, J. (2023). Fully integrated downstream process to enable next-generation manufacturing. *Biotechnology and Bioengineering*, 120, 1869–1881. https://doi.org/10.1002/bit.28384